Twin-Based Reinforcement Learning for Solving Multi-period Portfolio Optimization Problem

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Abstract. Portfolio optimization (PO) remains a key area of interest in finance. However, traditional methods face difficulties in creating an efficient and stable investment strategy due to the ever-changing nature of financial markets. These markets are influenced by internal factors that affect investment performance, such as volatility, trends, asset correlations, etc. As a result, a more dynamic and adaptive approach is needed to handle the market's volatility and uncertainty. This paper introduces a twin-based deep reinforcement learning framework to address these challenges. The framework consists of two agents: one manages the relaxed constraint problem, and the other handles the full constraint problem. The policies from the full constraint agent are periodically merged with those from the relaxed constraint agent, improving overall portfolio performance and risk management. We selected Deep Deterministic Policy Gradient (DDPG) as the core reinforcement learning algorithm. Additionally, we introduced constraints to mitigate market shocks during trading and prevent capital erosion. This study utilizes financial data from various Vietnamese stock classes, covering the period from January 2009 to December 2023. Fundamental stock price data is used to compute key metrics such as price change rates, cash dividends, and technical indicators like moving average convergence divergence (MACD). These comprehensive inputs boost model performance, while the interaction between the two modules further strengthens the framework through policy blending. Empirical experiments demonstrated that DDPG outperformed other well-known algorithms, including PPO, A2C, SAC, and TD3, while our twin-based approach improved overall portfolio performance and risk management.

Keywords: Reinforcement learning · Twin problems · Multi-period portfolio optimization problem · Portfolio risk awareness · Vietnam stocks

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1 Introduction

Portfolio optimization (PO) has been a central focus in finance since the introduction of Modern Portfolio Theory [1], which established the fundamental principles of risk-return trade-offs in portfolio investments. Advances in technology and data availability have since propelled the field, evolving from traditional optimization methods [2,3] to sophisticated deep learning techniques [4], with an increasing emphasis on reinforcement learning (RL) [5–9]. However, despite these innovations, current portfolio optimization models still face significant challenges.

First, conventional optimization models lack adaptability to the dynamic nature of financial markets. They rely heavily on historical data and make strong assumptions, such as normally distributed returns and static covariances, which often do not hold in real-world scenarios. As a result, models that continuously learn and adapt, like RL, are crucial for modern portfolio optimization. Second, existing models lack effective risk management strategies, especially for downside risk, and often fail to incorporate risk management indicators. Third, current models lack effective mechanisms for multi-period PO, which is essential for balancing short-term returns with long-term risks in high-volatility markets, where asset prices often experience significant and frequent fluctuations. In such markets, high volatility can offer opportunities for long-term investors to hedge against short-term fluctuations and potentially achieve significant rewards.

To address these challenges, we propose a twin-based RL framework that offers an adaptive, risk-aware solution for multi-period portfolio optimization in high-volatility contexts. The core innovation involves two distinct agents: one operating in a relaxed constraint environment to explore higher-risk strategies and maximize short-term returns, and the other adhering to full constraints to prioritize risk management and prevent capital erosion. These risk management constraints are designed to mitigate market shocks, making the framework robust in volatile markets. Additionally, we incorporate risk management indicators, i.e., Sortino ratio and moving average convergence divergence, into the RL framework for better risk management, such as downside risk. By periodically blending the policies of both agents, the model effectively balances return maximization with risk control, making it more suitable for volatile markets. The proposed framework is depicted in Fig. 1.

We train the twin-based RL framework using the Deep Deterministic Policy Gradient (DDPG) algorithm for its effectiveness in handling continuous action spaces, which is crucial for fine-tuning portfolio allocations. Our case study utilizes highly volatile historical Vietnamese stock data from 2009 to 2023. We demonstrate that our twin-based approach outperforms traditional RL algorithms, including Proximal Policy Optimization (PPO) [10], Advantage Actor-Critic (A2C) [11], Soft Actor-Critic (SAC) [12], and Twin Delayed Deep Deterministic Policy Gradient (TD3) [13], in terms of overall portfolio performance and risk-adjusted returns.

The structure of this paper is as follows: Sect. 2 describes the RL framework for multi-period PO, including the integration of risk-adjusted indicators.

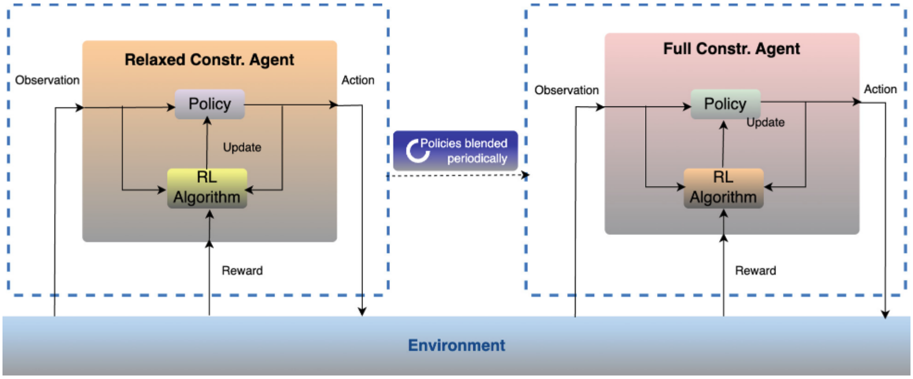


Fig. 1. The proposed twin-based deep RL framework.

Section 2 introduces the twin-based RL framework, designed to balance short-term returns and long-term risks under risk management constraints. Section 4 details the experimental results, and Sect. 5 discusses the study's limitations and future research directions.

2 RL Framework for Multi-Period PO

We address the multi-period PO problem, where a stock market investor seeks to generate returns over a given investment horizon, divided into several equal intervals, starting with fixed initial capital and multiple investment options. Our focus is on two main choices: dividend-paying stocks and a bank deposit with a fixed interest rate. The returns are influenced by stock price fluctuations, dividend payouts, and the fixed interest from the bank deposit. The investor's goal is to maximize returns while minimizing associated risks. Such a multi-period PO problem can be expressed as the Bellman expectation equation for policy π in the following RL framework, depicted in Eq. 1. Here, we use the Bellman expectation expression (without max) [14] because: i) *Complexity of solving for the optimal policy*: The Bellman optimality equation [15] demands that all possible actions be maximized in every state. For complex or continuous action spaces (such as adjusting fractional allocations in a portfolio), this maximization may be computationally expensive or even impractical. The Bellman expectation equation assesses a specific policy π, eliminating the need to solve an optimization problem at each step. ii) *Policy evaluation versus policy optimization*: Using the Bellman expectation equation for policy evaluation is simpler because it calculates the value of following a given policy without considering alternative actions. Optimizing policies using the Bellman optimality equation involves iteratively improving the policy, which can be computationally intensive in problems with large state or action spaces. iii) *Dynamic nature of financial markets*: PO involves a highly stochastic environment (market) that may not follow stationary transition probabilities, making it difficult to find a truly optimal policy.

However, a specific policy, such as one based on risk-adjusted returns (such as the Sharpe ratio), makes it easier to adapt and evaluate changing conditions. iv) *Practical constraints*: Many real-world constraints, such as transaction costs and stock trading volume limits, further complicate the optimization process.

$$Q^{\pi}(s_t, a_t) = \mathbb{E}\left[R(s_t, a_t) + \gamma Q^{\pi}(s_{t+1}, a_{t+1})\right] \tag{1}$$

where $Q^{\pi}(s_t, a_t)$ is the expected cumulative reward (or value) for taking action a_t in state s_t and following policy π thereafter; $r_t = R(s_t, a_t)$ is the reward received at time step t; γ is the discount factor, controlling the weight of future rewards (usually between 0 and 1); s_{t+1}, a_{t+1} presents the next state and action according to the environment and policy. The components of the RL frame are defined as follows:

- State s_t represents the current condition of the portfolio, including the investor's asset allocations, available capital, and stock prices. Each state consists of current stock prices and their moving average convergence divergence (MACD) [16] for providing insights into past trends to guide the agent's actions. The MACD is given by

$$EMA(t) = price(t) \times \frac{2}{N+1} + EMA(t-1) \times \left(1 - \frac{2}{N+1}\right) \tag{2}$$

  where N refers to the number of days. By incorporating MACD, agents can develop strategies that leverage the same trends that human traders use. For example, observing a MACD crossover might lead the agent to initiate a buy order, associating this action with future rewards if the price subsequently rises.
- Action a_t indicates the decisions the investor makes regarding allocating capital among the available investment options (stocks and bank deposits). Actions can involve long positions, short positions, or depositing funds.
- Reward r_t captures the investor's goal of maximizing returns, influenced by stock price changes, dividends, and fixed interest from the bank deposit. The reward is based on portfolio returns and incorporates the risk management indicator like the Sortino ratio [17], which accounts for downside risk, encouraging the agent to reduce negative volatility. It is given by

$$SortinoRatio = \frac{R_p - R_f}{\sigma_d} \tag{3}$$

  where R_p is the average portfolio return; R_f is the risk-free rate (the minimum acceptable return), and σ_d is the standard deviation of only the negative returns. A higher Sortino ratio signifies (i) a greater excess return relative to the amount of downside risk and (ii) smaller negative deviations of the returns, effectively balancing profitability while minimizing risk.
- Environments reflects the investment scenario in which investors operate, including factors such as stock prices, dividend distributions, and the fixed interest from the bank deposit.

This paper is carried out under the following assumptions: (i) short selling is not permitted; (ii) exchange and transaction fees are ignored; and (iii) there is no limit on the number of stocks that can be traded per transaction.

3 Twin-Based RL Framework for Multi-Period PO

We next extend the RL framework in Sect. 2 for a twin-based RL framework for the multi-period PO problem to balance short-term returns with long-term risks in high-volatility markets. Specifically, the twin-based framework operates with two distinct policies: one tailored for a relaxed environment to maximize short-term returns and another for a fully constrained environment to deal with long-term risks. Both policies are based on the standard Bellman optimality condition, but they differ in their reward structures and constraints.

3.1 Relaxed Policy

The relaxed model is primarily concerned with maximizing raw portfolio returns. The relaxed Bellman optimality condition is given by

$$Q^{\pi_r}(s_t, a_t) = \mathbb{E}\left[r_t^{\text{relaxed}} + \gamma^{\text{relaxed}} Q^{\pi_r}(s_{t+1}, a_{t+1})\right] \tag{4}$$

where π_r is the policy learned in the relaxed environment; γ^{relaxed} is the discount factor; r_t^{relaxed} is the immediate reward at step t based on pure returns (i.e., maximizing gains without considering risk constraints).

3.2 Constrained Policy

The constrained model aims to balance risk and return, considering risk-adjusted metrics like the Sortino ratio. The Bellman optimality condition is calculated as

$$Q^{\pi_f}(s_t, a_t) = \mathbb{E}\left[r_t^{\text{full}} + \gamma^{\text{full}} Q^{\pi_f}(s_{t+1}, a_{t+1})\right] \tag{5}$$

where π_f is the policy learned in the constrained environment; r_t^{full} is the reward for the constrained environment at time step t, which is a blended measure of risk-adjusted reward (like the Sortino ratio) and raw return; γ^{full} is the discount factor; and

$$r_t^{\text{full}} = (1-\alpha)\omega + \alpha\zeta \tag{6}$$

in which $\alpha \in [0, 1]$ is the risk-return weight; ω is the risk-adjusted reward; and ζ is the raw return; This guarantees that the agent's decisions consider not only maximizing returns but also the portfolio's risk (including volatility and downside risk). To avoid bias, we first normalize ζ and ω to a range of 0 to 1 before applying the risk-return weighting.

Two following constraints are considered. First, we integrated the "smooth transition constraint" as $|x_{i,t+1} - x_{i,t}| \leq \delta$ where $x_t = (\ldots, x_{i,t}, \ldots)$ is the vector of asset allocations (weights) at time t and $x_{i,t}$ represents the fraction of the portfolio allocated to asset i at time t. This constraint ensures that portfolio transitions are smooth, thereby reducing the risk of large market shocks and ensuring that no asset allocation changes by more than δ between consecutive time steps. Second, we define a termination threshold $\psi \in (0, 1)$, such that if the portfolio balance B_t drops to ψB_0, the episode is terminated early. The "early termination constraint" $B_t \leq \psi B_0$ indicates that the episode is stopped when the portfolio has lost more than $(1 - \psi) \times 100\%$ of its initial value, preventing further capital erosion, which mitigates the risk of incurring additional losses when the portfolio balance falls below a critical level.

3.3 Blended Policy

We define the blended policy π_b as a convex combination of the relaxed and constrained policies as follows:

$$\pi_b = \beta\pi_r + (1 - \beta)\pi_f \tag{7}$$

where $\beta \in [0, 1]$ is the blending factor. The blended policy π_b that results from mixing the relaxed and full-constraint policies has an effective Bellman equation that is a function of both reward structures. For the blended policy π_b, the Bellman equation is given by

$$Q^{\pi_b}(s_t, a_t) = \mathbb{E}\left[r_t^{\text{blended}} + \gamma^{\text{blended}} Q^{\pi_b}(s_{t+1}, a_{t+1})\right] \tag{8}$$

$$r_t^{\text{blended}} = (1 - \lambda) r_t^{\text{full}} + \lambda r_t^{\text{relaxed}} \tag{9}$$

where r_t^{blended} is the blended reward at time step t; $\lambda \in [0, 1]$ is the relaxed weight; γ^{blended} is the discount factor; $Q_{\text{blend}}(s_t, a_t)$ is the expected value for the blended policy, combining the constraints and risk considerations of the full-constraint environment and the return maximization of the relaxed environment. The reward is a weighted combination of the pure return r_t^{relaxed} and the constrained, risk-adjusted reward r_t^{full}. This blended approach essentially balances between exploiting high-return actions (from the relaxed policy) and adhering to risk-adjusted constraints (from the constrained policy).

4 Experiments

4.1 Dataset

This study collected data from 100 stocks listed on the Vietnam Stock Exchange over 15 years beginning in 2009[1], [2], [3] The dataset contains stock symbols, daily stock prices, and cash dividend payouts, with detailed information provided in Table 1. The data from 2009 to 2022 is used for training, while the 2023 data is set aside for testing.

[1] UPCoM stocks: https://www.hnx.vn/thong-tin-cong-bo-up-tcph.html.
[2] HNX stocks: https://www.hnx.vn/thong-tin-cong-bo-ny-tcph.html.
[3] HOSE stocks: https://www.hsx.vn.

Table 1. Dataset description

No.	Year	Stock number	Stock symbol	Transaction info	Company info	Index type	Dividend price/share (kVND)[a]
1	2009	100	Yes	Yes	Yes	Yes	Yes
2	2010	100	Yes	Yes	Yes	Yes	Yes
...	...	...	...	...	...	...	...
15	2023	100	Yes	Yes	Yes	Yes	Yes

[a] Note: 1 kVND is equivalent to 1,000 VND.

4.2 Parameters

In this study, we assess the investment performance over a one-year period, assuming the investor makes monthly decisions. The training phase spans 100,000 time steps. We use a bank deposit as the risk-free investment option, with a monthly risk-free rate of 0.45%. For calculating the MACD, we set the short, long, and signal windows to 12, 26, and 9, respectively. The relaxed weight λ is 0.3, and the initial capital is kVND 100,000. The policy is updated every 1,000 timesteps, with a blend factor β of 0.5. Early stopping occurs if there is a significant drop in the balance, defined as 30% of the initial investment. The $\max\_change$ allowed in the portfolio is 10%, and the risk-adjusted weight α is set to 0.5. Discount factors $\gamma^{\text{relaxed}} = \gamma^{\text{full}} = \gamma^{\text{blended}} = 0.99$, emphasizing a preference for future rewards. Both experiments were conducted on a 2022 MacBook Pro M2 running MacOS 15.0.1.

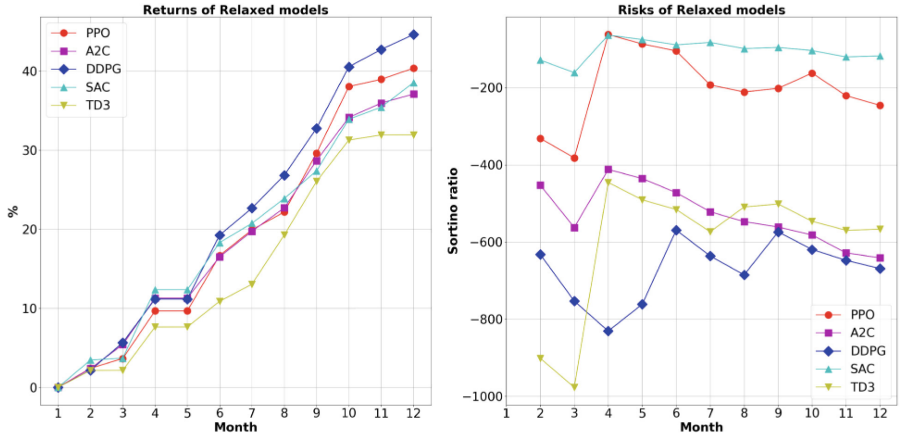
Window Size. Following additional practical experiments and backtesting, the window sizes have been determined as follows:

- For daily returns, start with a window size of 30–60 steps. This covers approximately 1–2 months, providing a better understanding of how the portfolio handles volatility over a more meaningful period.
- For weekly returns, consider a window size of 10–12 steps to cover about 2–3 months.
- For monthly returns, a window size of 6–12 steps (representing 6 months to 1 year) should give you a good view of downside risk.

In this experiment, we evaluate the investment over one year, making monthly decisions, with a window size of 6 selected for analysis.

4.3 Results

In this experiment, we will evaluate our proposed solution for addressing the multi-period portfolio optimization problem. To facilitate comparison, we will test our twin-based approach alongside two other methods. One approach involves solving the relaxed constraint problem, disregarding investment risks but ensuring the investment plan still functions properly. The other approach tackles the full problem, maintaining risk considerations and strict constraints,

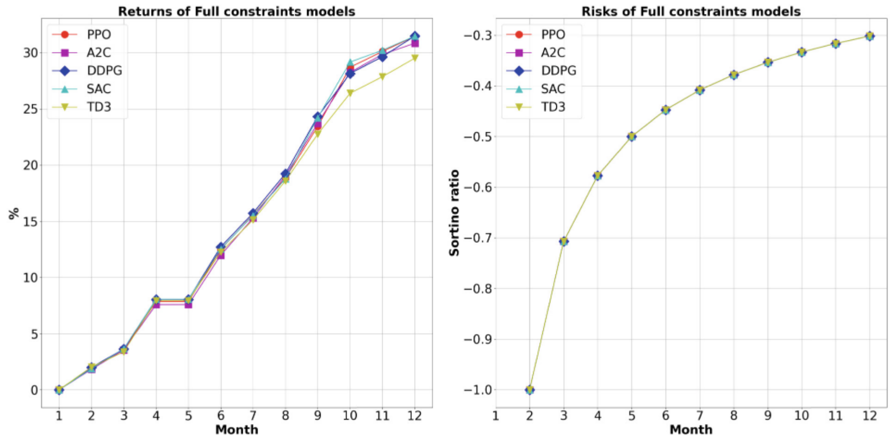


(a) Profits of the relaxed constraint issue. (b) Risks of the relaxed constraint problem.

Fig. 2. The returns and risks of the relaxed constraint problem do not aim to minimize risks.

but unlike the twin-based approach, it does not incorporate relaxed policies during training.

As depicted in Fig. 2a, Fig. 3a, and Fig. 4a, it is clear that the relaxed-based DRL solutions, which do not incorporate risk management, achieve higher returns compared to the other two approaches that do account for risk. Among these, DDPG consistently provides the highest returns in each category (over 40%, 30%, and 35% returns, respectively). However, when comparing the full



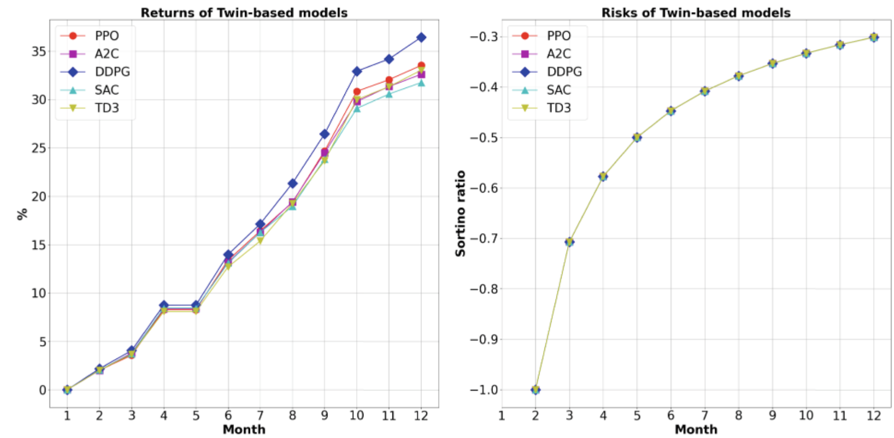
(a) Profits of the full constraint problem. (b) Risks of the full constraint problem.

Fig. 3. Profits and Sortino ratio values of the full constraint problem are evaluated without integrating relaxed environments during the training process.

constraint problem with and without the integration of the relaxed environment, the version that incorporates the relaxed environment delivers better returns. In terms of risk management, the relaxed constraint problem performs the worst due to the absence of risk considerations, making it less favorable compared to the full constraint problem (both with and without relaxed policies). Additionally, because among PPO, A2C, DDPG, SAC, and TD3, DDPG generates the highest returns in the relaxed constraint problem, it also incurs the highest risks, often dropping below −600, as seen in Fig. 2b. Therefore, in the absence of risk management, higher returns come with significantly higher risks.

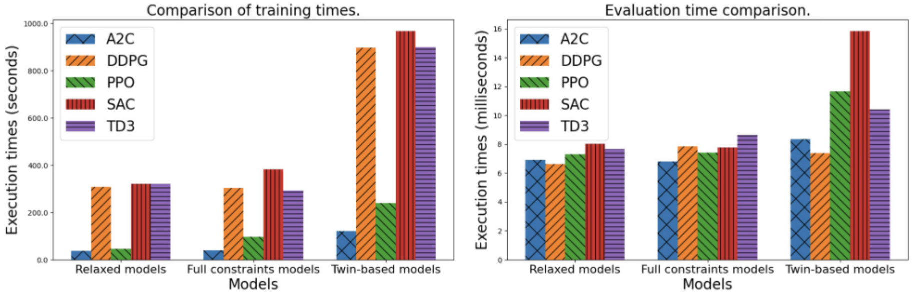
Looking more closely at the full constraint problem, which focuses on risk management, incorporating the relaxed environment leads to improved returns. This shows the increase in returns, with returns exceeding 35% compared to over 30% without the relaxed environment, as shown in Fig. 4a and Fig. 3a. While the twin-based approach offers higher returns, the associated risks remain nearly the same as in the full constraint problem without the relaxed environment, as illustrated in Fig. 4b and Fig. 3b. It is also important to note that all RL models (PPO, A2C, DDPG, SAC, and TD3) exhibit similar risk values. However, when comparing the risks of these two approaches to those of the relaxed constraint approach, the full constraint approaches are much better. For example, the full constraint problems (with or without the relaxed environment) reach a maximum Sortino ratio of –0.3, which is significantly better than the relaxed approach's best case (SAC) with a Sortino ratio of –200, as shown in Fig. 2b. In conclusion, for risk-averse investors, we recommend using the twin-based framework with DDPG, as it provides higher returns and better risk management.

Regarding execution times, training our proposed twin-based approach takes over twice as long as the relaxed constraint problem and more than double the



(a) Profits of the full constraint problem. (b) Risks for the full constraint problem.

Fig. 4. The returns and risks of the full constraint problem are assessed using our proposed Twin-based approach.



(a) Training times.

(b) Evaluation times.

Fig. 5. The execution times for the training and evaluation phases of the relaxed constraint problem, the full constraint problem without using our twin-based solution, and the full constraint problem with the twin-based solution.

time of the full constraint problem without the relaxed environment. The longest training time is for SAC, around 900 s, followed by DDPG at approximately 850 s, as shown in Fig. 5a. For the evaluation phase, both are relatively quick, with the longest evaluations for the twin-based solution taking less than 16 ms, as shown in Fig. 5b.

5 Conclusion

In the past, before selecting an investment plan, investors had to consider various factors such as stock prices, cash dividends, and technical analysis. Their goal is to maximize returns while minimizing investment risk, but determining the best timing and frequency for trades was both difficult and time-consuming. To address this challenge, we propose a multi-agent adaptive portfolio trading framework based on DRL, incorporating a twin-based approach with DDPG as the core algorithm.

Our framework utilizes a twin-problem approach integrated with reinforcement learning. Two agents are deployed: one addresses the relaxed constraint problem, while the other handles the full constraint problem. The policies from the full constraint agent are periodically merged with those of the relaxed constraint agent, leading to improved overall portfolio performance and risk management, as measured by the Sortino ratio. Additional constraints are introduced to mitigate market shocks, particularly when investors hold significant capital relative to the market, and to prevent capital erosion. In the paper, we also discuss the reasons for selecting DDPG as the core reinforcement learning algorithm and conduct additional comparisons with PPO, A2C, SAC, and TD3 during the experiments.

This study draws on financial datasets from various Vietnam stock classes, covering the period from January 2009 to December 2023. Fundamental stock price data is used to calculate key metrics such as price change rates, cash dividends, and technical indicators like MACD. These comprehensive inputs improve model performance, and the interaction between the two modules strengthens the framework through the blending of policies.

In this research, we refine and further develop reinforcement learning algorithms based on the twin-based approach to address multi-period portfolio management with constraints. Future work could explore quantitative trading experiments that account for limited transaction volumes and offer further comparison standards with models like Markowitz's mean-variance, integrating time series techniques for stock price forecasting and transaction volume limits. This could ultimately lead to a fully automated trading system for practical application. Theoretically, our proposed model appears adaptable to any stock market and other application domains, as it does not impose strictly defined constraints. Therefore, future research could explore applying our solution to additional markets and domains, and further experiments would be essential for validation.

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